

Question	Answer	Mark	Mark scheme	Additional guidance
29	No Supported	P1	for finding the area of a circle eg $\pi \times 0.8^2$ (= 2.01...)	Must be area of circle and not part of a volume, eg $\pi r^2 h$ May be seen as $2\pi r^2$
		P1	for finding the curved surface area eg $2\pi \times 0.8 \times 1.8$ (= 9.047...)	May be seen from $2\pi r h$ or from $\pi d h$
		P1	for use of the coverage information with an area eg “2.01...” $\div 5$ (= 0.402...) or “4.02...” $\div 5$ (= 0.804...) or “9.047...” $\div 5$ (= 1.8095...) or “11.058” $\div 5$ (= 2.2116.. or “13.069...” $\div 5$ (= 2.6138...) OR for process to find total coverage for comparison eg 5×7 (= 35)	Accept numbers without working written to no less than 2dp Do not award if a volume has been used as part of the calculation. An independent mark for 5×7
		P1	(dep P1) for finding total surface area for 3 tanks eg [total surface area] $\times 3$ (= 39.2...) OR for complete process to find the number of tins needed for total area of 3 tanks eg “13.069”.... $\times 3 \div 5$ (= 7.84.....) OR for complete process to find coverage needed from each tin eg “13.069”... $\times 3 \div 7$ (= 5.6...)	[total surface area] must come from the addition of two attempts at area, but not from volume.
		C1	for conclusion “No” supported by accurate figures eg 8 tins or 7.84 (> 7) or 39.2 > 35 or 5.6 (> 5)	Clear statement that there is not enough paint supported by correct figures for comparison. NB: $2.6 \times 3 = 9$ tins needed is inaccurate 8 or 7.84 tins is sufficient without restating the 7, 5.6 m ² is sufficient without restating the 5 but 39.2 and 35 are needed for comparison. A statement of “No, 8 tins” alone gets 0 marks without supporting working.